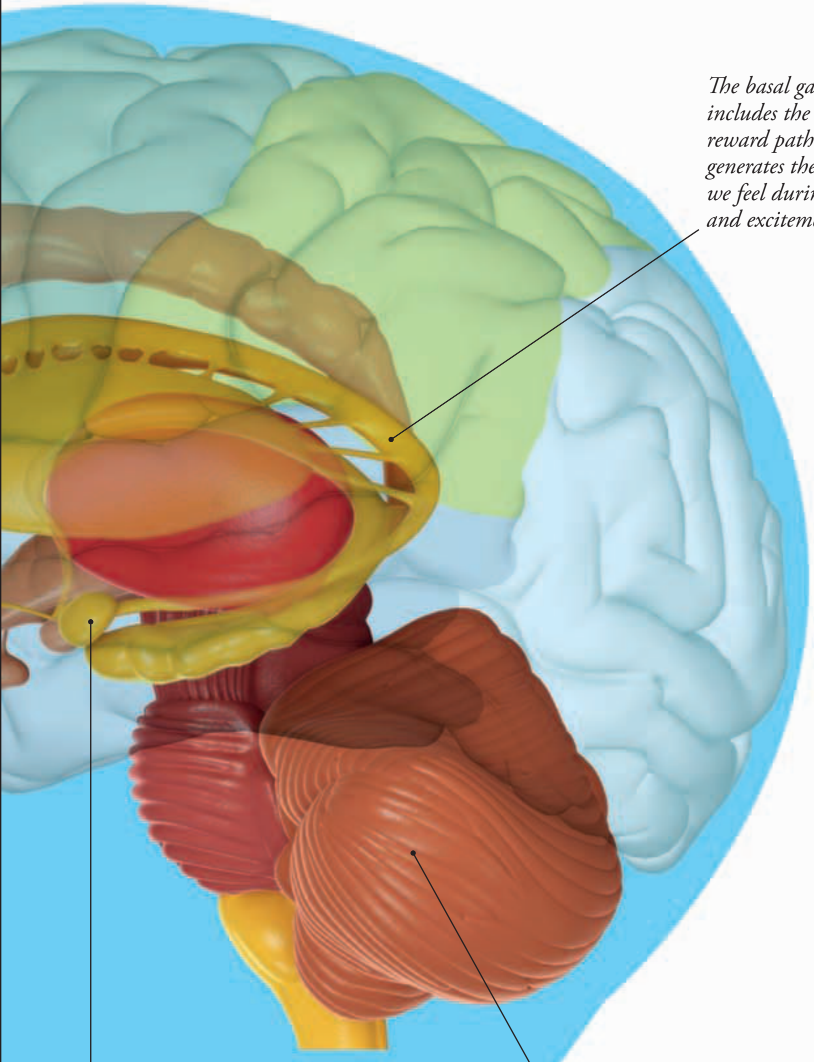
THE MATURING BRAIN

These scans show how the brain's surface tissue (gray matter) changes during the teenage years. Red and yellow areas have a large amount of gray matter, whereas blue areas have less. As unused brain circuits are pruned away, the amount of gray matter falls. Although the remaining gray matter has less potential to learn new skills, it becomes much more efficient in the skills it has acquired.

13 years old | 15 years old | 18 years old



The basal ganglia region includes the brain's reward pathway. It generates the "buzz" we feel during pleasure and excitement.

The amygdala is the emotional hub of the brain. It creates powerful feelings such as fear and anger. It is one of the areas of the brain that makes teenagers moody.

The cerebellum helps coordinate the body's movements. During the teenage years, when the body grows rapidly, the cerebellum has to relearn how to control our movements.

RECKLESS YOUTH

People take the most risks at the age of 14. At this age, the basal ganglia—the parts of the brain that generate the thrill of excitement—are fully formed. But the thinking frontal lobes are not, so the brakes are missing. Lacking sound judgment, some teenagers take crazy risks, especially when trying to impress friends.

MOODY PHASE

Tantrums and mood swings are common among teenagers. One reason is that the parts of the brain that create emotions develop faster than the parts that help us control emotions. Adults are good at suppressing or hiding their feelings and staying polite, but teenagers are more likely to act first and think later.